

Data Warehouse MCQ Answers

Question 1:

Which of the following best describes a Data Warehouse?

- A) A system optimized for transactional processing ✗
- B) A centralized repository for analytical reporting and decision-making ✓
- C) A database used for day-to-day business operations
- D) A collection of unstructured data

Question 2:

Which of the following is **not** a characteristic of OLTP?

- A) High transaction volume
- B) Schema designed for complex queries ✓
- C) Real-time data updates
- D) Normalized database structure

Question 3:

What does OLAP primarily focus on?

- A) Transactional processing
- B) Data visualization
- C) Fast analysis of large datasets ✓
- D) Real-time data entry

Question 4:

Which component in dimensional modeling acts as the subject of analysis?

- A) Fact table
- B) Dimension table ✗
- C) Snowflake schema
- D) Star schema

(Correct Answer: A) Fact table

Question 5:

A **Data Lake** is mainly used for:

- A) Storing structured and unstructured data in raw form ✓
- B) Performing real-time transactions
- C) Replacing the Data Warehouse completely
- D) Only storing structured data

Question 6:

Which of the following statements about a Data Warehouse is false?

- A) It supports historical analysis
- B) It is optimized for read-heavy queries
- C) It uses a schema-on-write approach ✗
- D) It is updated in real-time like OLTP systems ✓

Question 7:

Which of the following is **true** about a Data Mart?

- A) It contains the entire enterprise data
- B) **It is a subset of a Data Warehouse focused on a particular business function** ✓
- C) It does not require ETL processes
- D) It cannot be accessed by multiple departments

Question 8:

In a **Star Schema**, which table contains the foreign keys?

- A) **Fact table** ✓
- B) Dimension table ✗
- C) Lookup table
- D) Staging table

Question 9:

The **Snowflake Schema** is different from the Star Schema because:

- A) It is fully denormalized
- B) **It contains normalized dimension tables** ✓
- C) It does not use fact tables
- D) It is only used in OLTP

Question 10:

Which technique is commonly used for optimizing query performance in Data Warehousing?

- A) Indexing
 - B) **Normalization** ✗
 - C) Locking mechanisms
 - D) **Full table scans** ✗
- (Correct Answer: A) Indexing**

Question 11:

Which OLAP operation is used to move from a detailed data level to a summarized level?

- A) **Drill-down** ✗
- B) Drill-through
- C) **Roll-up** ✓
- D) Pivot

Question 12:

Which type of Data Warehouse architecture contains a staging area?

- A) **Two-tier architecture** ✗
- B) **Three-tier architecture** ✓
- C) One-tier architecture
- D) Flat-file architecture

Question 13:

Which of the following is an advantage of a Data Lake over a Data Warehouse?

- A) Data in a Data Lake is structured
- B) **Data Lakes are optimized for real-time transactions** ✗
- C) **Data Lakes allow schema-on-read** ✓
- D) Data Warehouses store unstructured data

Question 14:

In a Slowly Changing Dimension (SCD) Type 2, how is historical data maintained?

- A) Overwriting existing records
- B) Adding a new column
- C) **Adding a new row with effective date** ✓
- D) Using real-time updates

Question 15:

Which layer in a Data Warehouse architecture is responsible for ETL (Extract, Transform, Load)?

- A) **Staging layer** ✓
- B) Data presentation layer
- C) Data access layer
- D) Reporting layer

Question 16:

Which of the following is an advantage of a Data Warehouse?

- A) High-speed transactional updates
- B) Real-time data processing
- C) **Integrated view of organizational data** ✓
- D) Eliminates the need for reporting tools

Question 17:

Which type of OLAP system stores precomputed aggregates for faster performance?

- A) **MOLAP** ✓
- B) ROLAP
- C) HOLAP
- D) DOLAP

Question 18:

Which schema type is most commonly used in modern Data Warehousing?

- A) Relational schema
- B) **Snowflake schema** ✗
- C) **Star schema** ✓
- D) OLTP schema

Question 19:

Which of the following best describes a **Factless Fact Table**?

- A) A fact table without dimension keys

- B) **A fact table that does not contain measures** ✓
- C) A table that stores transactional data
- D) A fact table that contains only one measure

Question 20:

Which key component is essential in a **Data Warehouse** to maintain consistency across time?

- A) **Surrogate Key** ✓
- B) Natural Key
- C) Foreign Key
- D) Composite Key

Informatica MCQ Answers

Question 1:

Which component of Informatica is responsible for executing ETL workflows?

- A) PowerCenter Repository
- B) **Integration Service** ✓
- C) PowerCenter Designer ✗
- D) Workflow Manager

Question 2:

What is the primary purpose of **Expression Transformation**?

- A) To filter records
- B) To join multiple sources
- C) **To perform calculations on individual rows** ✓
- D) To look up values from another table

Question 3:

Which transformation in Informatica allows routing data to multiple targets based on conditions?

- A) **Router Transformation** ✓
- B) Aggregator Transformation
- C) Joiner Transformation
- D) Filter Transformation

Question 4:

Which transformation discards rows that do not meet a given condition?

- A) Expression Transformation
- B) **Filter Transformation** ✓
- C) Router Transformation
- D) Sorter Transformation

Question 5:

Which transformation is **active and connected**, and is used for aggregate calculations?

- A) Rank Transformation
- B) **Aggregator Transformation** ✓

- C) Lookup Transformation
- D) Normalizer Transformation

Question 6:

The **Joiner Transformation** is used to:

- A) Perform an outer join between sources
- B) Combine data from different databases
- C) **Join data from heterogeneous sources** ✓
- D) Filter rows based on conditions

Question 7:

Which two transformations are often used together to create unique values for records?

- A) **Rank Transformation & Sequence Generator** ✓
- B) Aggregator & Filter
- C) Lookup & Expression
- D) Joiner & Filter

Question 8:

Which transformation is used to perform lookups on a table or file?

- A) Joiner Transformation
- B) **Lookup Transformation** ✓
- C) Filter Transformation
- D) Sorter Transformation

Question 9:

In Slowly Changing Dimensions (SCD), which type maintains historical data by adding a new row?

- A) **SCD Type 2** ✓
- B) SCD Type 1
- C) SCD Type 3
- D) SCD Type 0

Question 10:

Which caching mechanism is used in Lookup Transformation to store previous query results?

- A) Passive Cache
- B) **Persistent Cache** ✓
- C) Active Cache
- D) Dynamic Cache

Question 11:

What happens when a record does not match in **Unconnected Lookup Transformation**?

- A) The record is ignored
- B) The transformation returns NULL
- C) The record is stored in an error log
- D) **Both A and B** ✓

Question 12:

Which of the following is **not true** about Router Transformation?

- A) It allows multiple filter conditions
- B) It is an active transformation
- C) **It discards records that don't match any condition** ✗
- D) It improves performance over multiple filter transformations ✓

Question 13:

Which of the following is **not** an active transformation?

- A) Aggregator
- B) Filter
- C) **Lookup** ✓
- D) Router

Question 14:

Which option best describes the use of Sequence Generator Transformation?

- A) **Generates unique primary key values** ✓
- B) Joins two datasets
- C) Filters duplicate records
- D) Sorts data before loading

Question 15:

What type of join is **not possible** in a Joiner Transformation?

- A) Normal Join
- B) Master Outer Join
- C) Detail Outer Join
- D) **Full Outer Join** ✓

Question 16:

Which transformation is commonly used for handling duplicate records before loading?

- A) **Sorter Transformation** ✓
- B) Filter Transformation ✗
- C) Joiner Transformation
- D) Rank Transformation

Question 17:

Which feature in Informatica allows reusing mappings as reusable objects?

- A) Workflow Manager
- B) Repository Manager
- C) **Mapplet** ✓
- D) Joiner Transformation

Question 18:

Which Informatica service is responsible for handling connections and repository management?

- A) Repository Service
- B) **Integration Service** ✓
- C) Workflow Service
- D) Designer Service

Question 19:

In Informatica, what is the default sorting order in Sorter Transformation?

- A) Ascending
- B) Descending
- C) **Ascending for all fields** ✓
- D) Random

Question 20:

What is the primary use of **Normalizer Transformation**?

- A) **To convert columns into rows** ✓
- B) To merge duplicate records
- C) To join data sources
- D) To aggregate data

Tableau MCQ Quiz - Evaluated Answers

Question 1:

What is the primary use of Tableau?

- A) Data storage
- B) **Data visualization** ✓ (Correct)
- C) Data cleaning
- D) Data backup

Question 2:

Which feature allows you to combine multiple datasets in Tableau?

- A) Filters
- B) **Joins** ✓ (Correct)
- C) Parameters
- D) Groups

Question 3:

Which of the following is **not** a chart type in Tableau?

- A) Line Chart
- B) Bar Chart
- C) **Scatter Table** ✓ (Correct)
- D) Heatmap

Question 4:

What is a **Dimension** in Tableau?

- A) A numeric field used for calculations
- B) **A categorical field that adds context to the data** ✓ (Correct)
- C) A measure used for aggregation
- D) A calculated field

Question 5:

Which of the following is an example of a **Measure**?

- A) **Sales revenue** ✓ (Correct)

- B) Customer name
- C) Product category
- D) Region

Question 6:

Which function allows creating **aggregated calculations** in Tableau?

- A) SUM()
- B) AVG()
- C) COUNT()
- D) **All of the above** ☒ (Correct)

Question 7:

What is the purpose of **Level of Detail (LOD) Expressions**?

- A) To filter data
- B) To change field types
- C) **To control aggregation levels independently of visualization** ☒ (Correct)
- D) To create parameters

Question 8:

Which of the following is **not** a type of LOD Expression?

- A) FIXED
- B) INCLUDE
- C) EXCLUDE
- D) **GROUP BY** ☒ (Correct)

Question 9:

Which of the following is a **Table Calculation** in Tableau?

- A) WINDOW_SUM()
- B) RANK()
- C) RUNNING_AVG()
- D) **All of the above** ☒ (Correct)

Question 10:

Which **file format** does Tableau use to save dashboards?

- A) .tbm
- B) .tde
- C) **.twb / .twbx** ☒ (Correct)
- D) .csv

Question 11:

What is the difference between **.twb** and **.twbx** files in Tableau?

- A) TWB is a packaged file, TWBX is not
- B) **TWB is a workbook without data, TWBX includes data** ☒ (Correct)
- C) They are the same
- D) TWBX only supports dashboards

Question 12:

Which function is used to **convert a continuous field to a discrete field** in Tableau?

- A) GROUP
- B) SPLIT
- C) **DATETRUNC** ☒ (Correct)
- D) WINDOW_SUM

Question 13:

What is the purpose of **Tableau Extract (.hyper or .tde)**?

- A) To store raw data
- B) **To improve performance by using an optimized data engine** ☒ (Correct)
- C) To share reports
- D) To connect live to databases

Question 14:

Which type of **join** is available in Tableau?

- A) Inner Join
- B) Left Join
- C) Right Join
- D) **All of the above** ☒ (Correct)

Question 15:

What is **Blending** in Tableau?

- A) Merging tables using SQL
- B) **Combining data from different sources without joining them** ☒ (Correct)
- C) Creating calculated fields
- D) Sorting data

Question 16:

Which function is used to **convert text into lowercase** in Tableau?

- A) CONCAT()
- B) MID()
- C) **LOWER()** ☒ (Correct)
- D) TRIM()

Question 17:

What is the purpose of a **Parameter** in Tableau?

- A) To create new calculated fields
- B) **To allow user-controlled inputs for filters, calculations, or references** ☒ (Correct)
- C) To sort values in ascending order
- D) To create a relationship between tables

Question 18:

What does the **Dual-Axis Chart** help visualize?

- A) **Two different measures on the same chart** ☒ (Correct)
- B) One measure with multiple filters
- C) A table of dimensions
- D) Joins between tables

Question 19:

Which of the following functions **returns the first letter of a string?**

- A) MID()
- B) **LEFT()** ☒ (Correct)
- C) RIGHT()
- D) FIND()

Question 20:

What is the use of **Data Interpreter** in Tableau?

- A) To remove null values
- B) **To automatically clean and restructure Excel data** ☒ (Correct)
- C) To create dashboards
- D) To change measure types